

Lab 1 - Requirements Table

Problem Statement 07: Hostel Maintenance & Issue Ticketing System

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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall let a resident raise a maintenance ticket with the room number, issue type (plumbing, electrical or carpentry), severity and up to five photos.	High	Pass: ticket is saved with a unique ID and shows up in the warden queue. Fail: form is accepted with the room number or issue type left blank.	This is the main entry point. If the details are not captured properly the ticket cannot be routed.	Mention the photo size limit (5 MB) so the criteria can be tested.
FR-002	Functional	The system shall let the warden assign an open ticket to a maintenance staff member and set a target date for completion.	High	Pass: status changes from Open to Assigned, staff name and target date are stored, and the staff member gets a notification. Fail: ticket can be assigned without picking a staff member.	Assignment is what actually gets the work scheduled and starts the SLA timer.	Should we allow reassignment? Not covered.
FR-003	Functional	The system shall let the assigned staff member change the ticket status to In Progress, On Hold or Resolved, and enter a short remark each time.	High	Pass: every status change is logged with time, staff ID and remark, and the resident can see the new status. Fail: status is changed with the remark left empty.	Residents need to see progress, and these timestamps are what SLA tracking uses.	Fine. Maybe fix a minimum length for the remark.
FR-004	Functional	The system shall not allow a ticket to be closed until the resident who raised it has either accepted or rejected the repair.	High	Pass: a Resolved ticket moves to Closed only after the resident accepts. If rejected it goes back to In Progress and the warden is told. Fail: ticket gets closed with no response from the resident.	The problem statement asks for resident sign-off. Stops work being closed before it is actually done.	What if the resident never replies? Needs an auto-close rule.
FR-005	Functional	The system shall let residents, staff and wardens search past tickets by date, status, hostel block and issue type.	Medium	Pass: search returns the correct tickets, and a resident only sees their own. Fail: a resident is able to open someone else's ticket.	Useful for spotting repeat faults in the same room or block, and for audit.	Add a response time here so it is measurable.
NFR-001	Nonfunc. (Performance)	The system shall send an automatic SMS and email alert to the warden if a high-priority ticket has had no status change for 24 hours.	High	Pass: in a load test with 200 open tickets, every ticket crossing 24 hours triggers one alert, and the alert is logged. Fail: the 24 hours pass with no alert, or the same ticket alerts twice.	Without this, critical complaints can sit in the queue with nobody noticing.	What happens if the SMS gateway is down?
NFR-002	Nonfunc. (Security)	The system shall restrict actions by role, so residents, staff and wardens can only do what their role allows. All traffic must be over HTTPS.	High	Pass: a resident calling the assign-ticket endpoint is blocked and the attempt is logged. No page loads over plain HTTP. Fail: any request meant for another role goes through.	Tickets contain room numbers and photos of rooms, so access has to be controlled.	Session timeout could be added here.

Traceability (which requirement maps to which use case)

Req ID	Use Case	Notes
FR-001	UC-01 Log Maintenance Ticket	Uses <<include>> Validate Ticket Details
FR-002	UC-03 Assign Maintenance Staff	Warden
FR-003	UC-06 Update Ticket Status	Maintenance staff
FR-004	UC-02 Review Resolution & Signoff	Resident
FR-005	UC-05 View Ticket History	All three actors
NFR-001	UC-04 Track SLA, Send Escalation Alert	Shown using <<extend>>
NFR-002	Applies to all use cases	Not drawn separately in the diagram

Actors: Hostel Resident, Maintenance Staff, Maintenance Warden, and Notification Service as a supporting actor.

Use cases: UC-01 Log Maintenance Ticket, UC-02 Review Resolution & Signoff, UC-03 Assign Maintenance Staff, UC-04 Track SLA, UC-05 View Ticket History, UC-06 Update Ticket Status. Validate Ticket Details and Send Escalation Alert are the included and extending use cases.